

SmartBOQ Pro

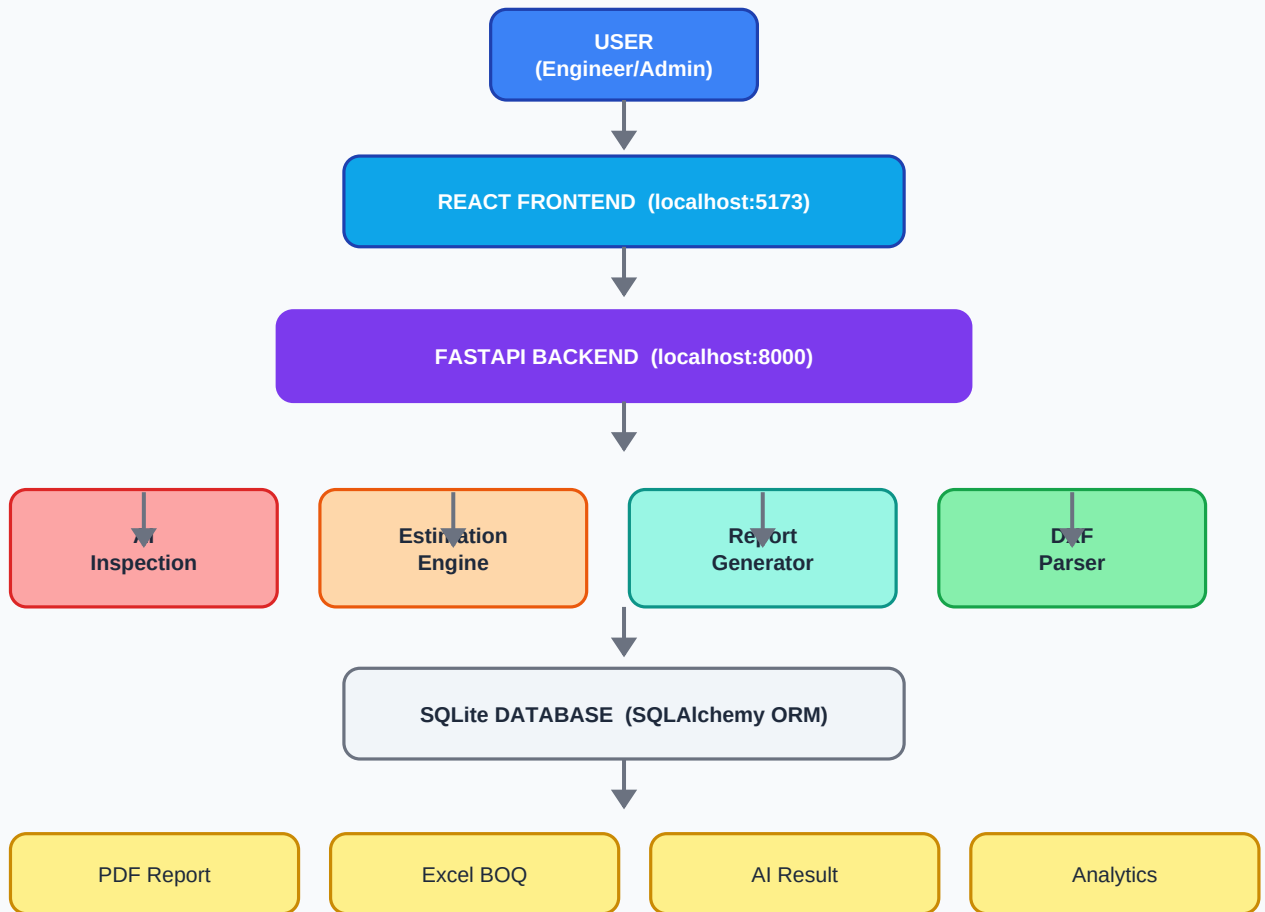
Feature Flowcharts & Diagrams

Complete visual guide: Input → Process → Output for every feature

System Overview — All Modules at a Glance

SmartBOQ Pro — Complete System Overview

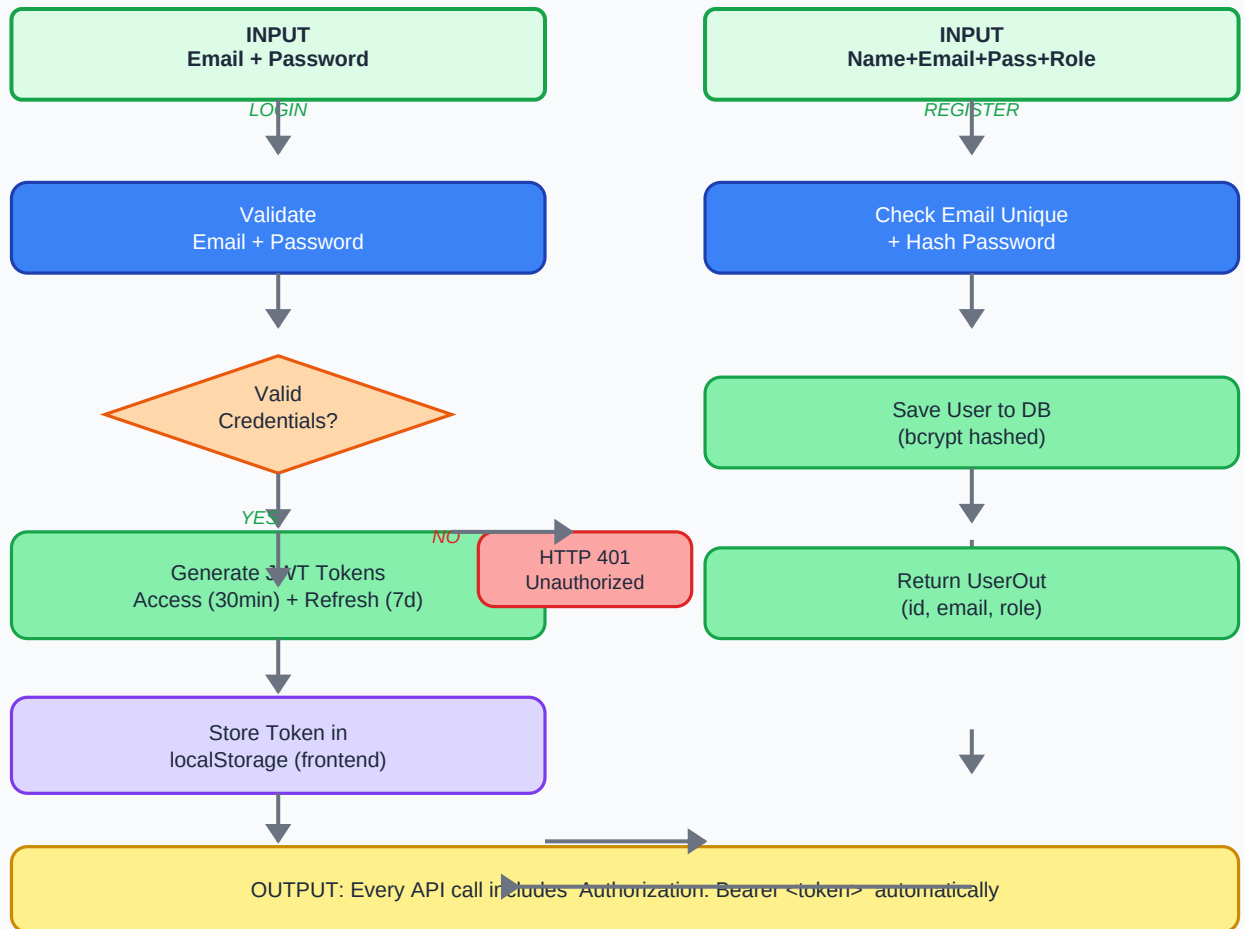
INPUT → PROCESS → OUTPUT for every module



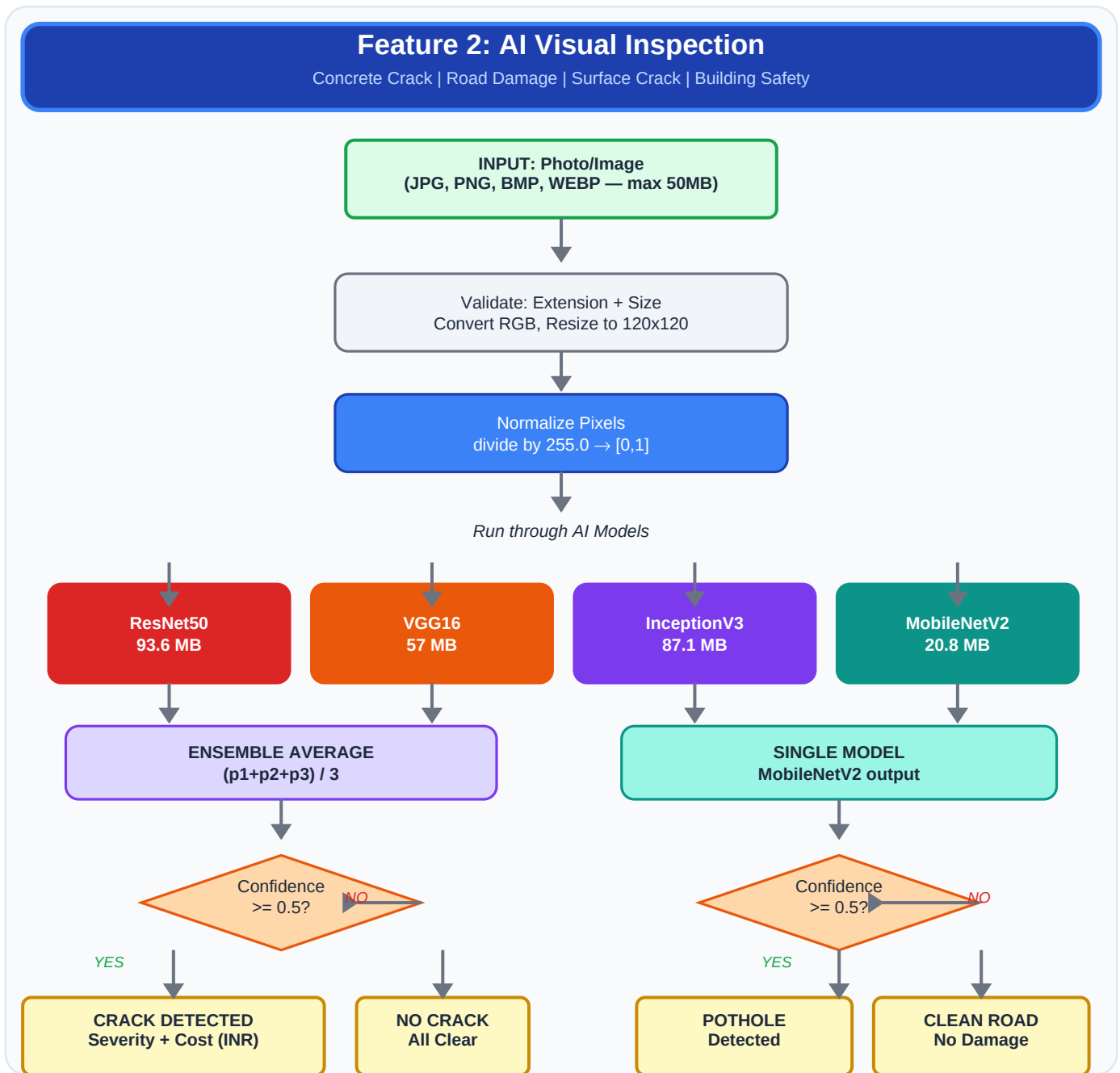
Feature 1: Authentication (Login/Register/JWT)

Feature 1: Authentication Flow

Login / Register / JWT Token Management



Feature 2: AI Visual Inspection (4 Model Types)



AI Inspection: Severity Scoring & Cost Output

AI Inspection — Severity Scoring & Output

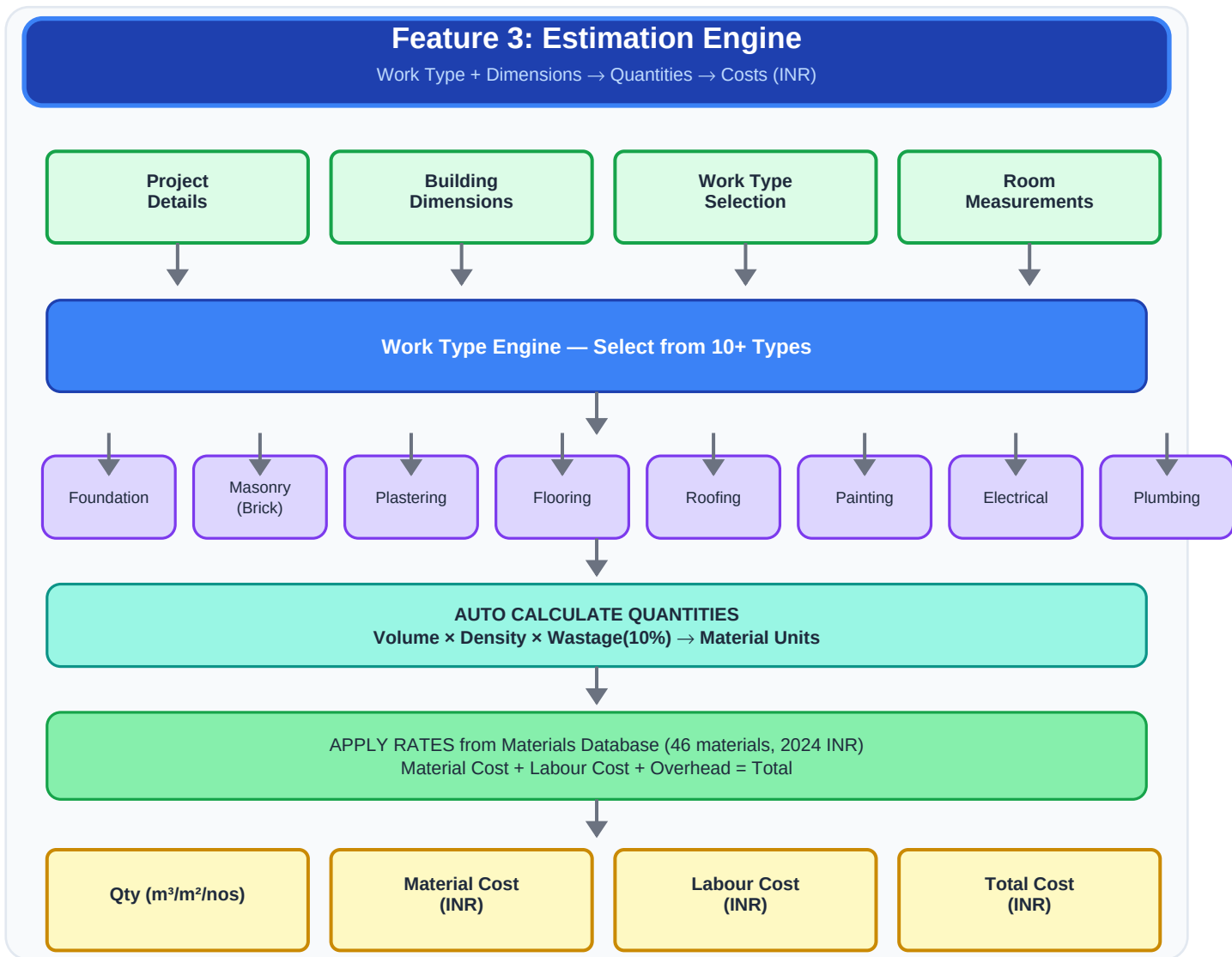
Confidence Score → Severity → Repair Cost (INR)

Confidence	Severity	Urgency	Repair Cost (INR)	Color
>= 0.90	CRITICAL	Immediate repair	■50,000 – ■2,00,000	
>= 0.75	HIGH	Within 7 days	■20,000 – ■80,000	
>= 0.55	MODERATE	Within 30 days	■5,000 – ■25,000	
>= 0.50	LOW	Monitor regularly	■1,000 – ■8,000	

FINAL OUTPUT stored in Database & shown on screen:

Inspection ID | Type | Model Used | Confidence %
Severity Level | Recommendation | Repair Cost Range | Timestamp

Feature 3: Estimation Engine



Feature 4: BOQ Generator + Export

Feature 4: Bill of Quantities (BOQ) Generator

Estimates → BOQ → PDF/Excel Export

INPUT: Saved Estimates
(from Estimation Engine)

INPUT: Profit% + Contingency%
(default: 10% + 5%)

Generate BOQ Line Items
(Item No. | Description | Unit | Qty | Rate | Amount)

Calculate Totals
Sub-Total + Profit Amount + Contingency = Grand Total (INR)

Save BOQ to Database (linked to Project ID)

Export PDF
(ReportLab — formatted tables)

Export Excel
(OpenPyXL — .xlsx file)

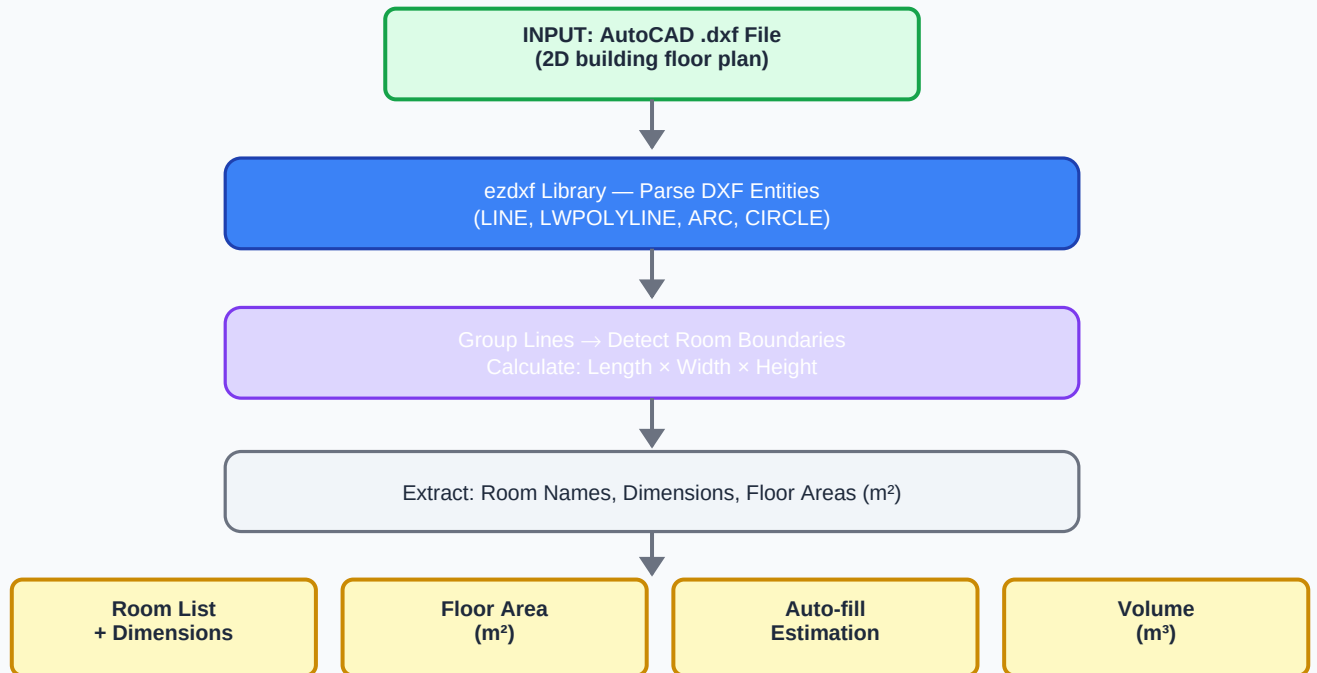
View on Screen
(React table with pagination)

OUTPUT: Professional BOQ Document | Grand Total (INR) | Ready for Client/Tender

Feature 5: AutoCAD DXF Parser

Feature 5: AutoCAD DXF File Parser

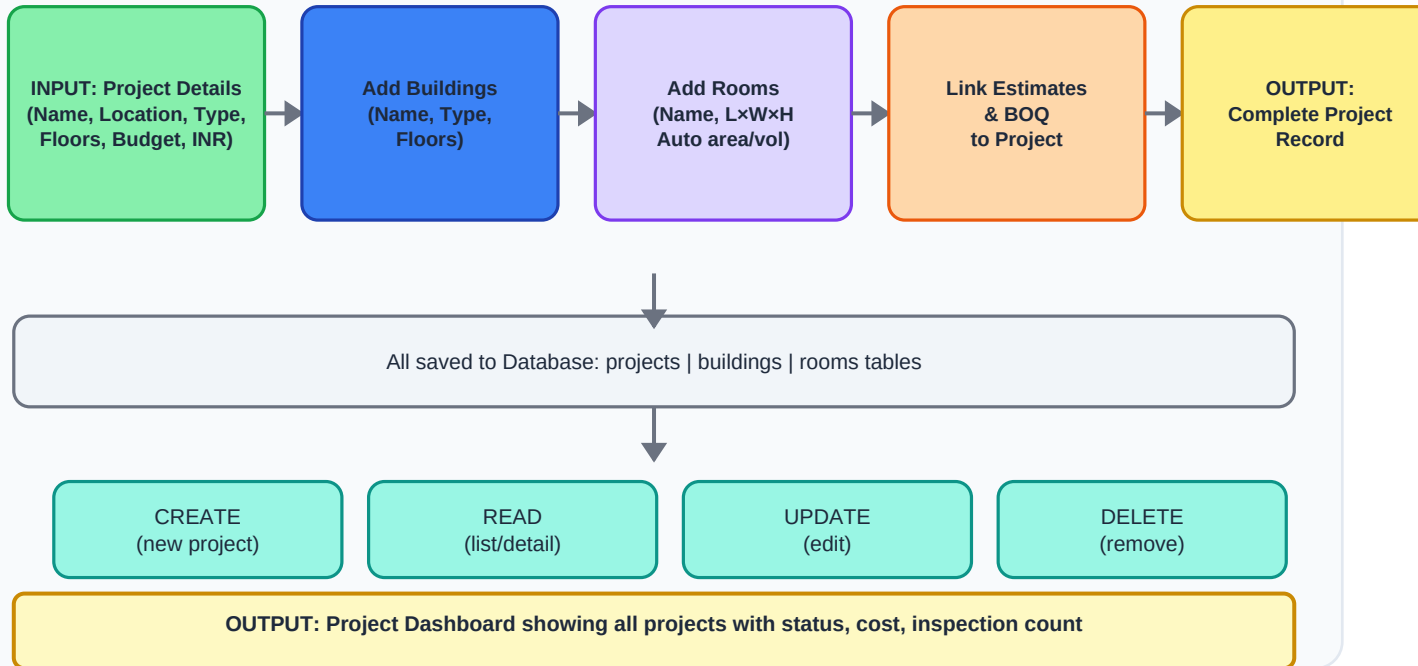
Upload .dxf Drawing → Auto-Extract Room Dimensions



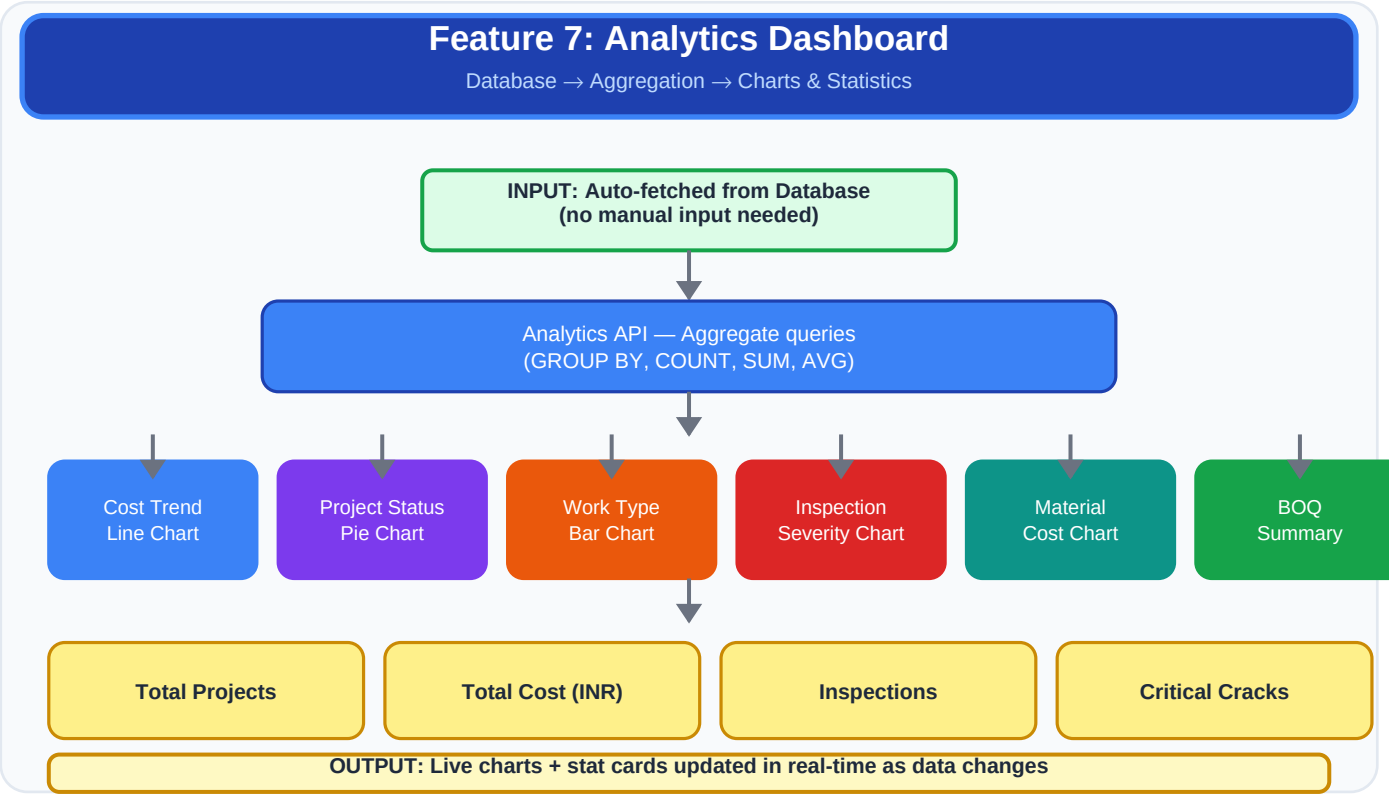
Feature 6: Project & Building Management

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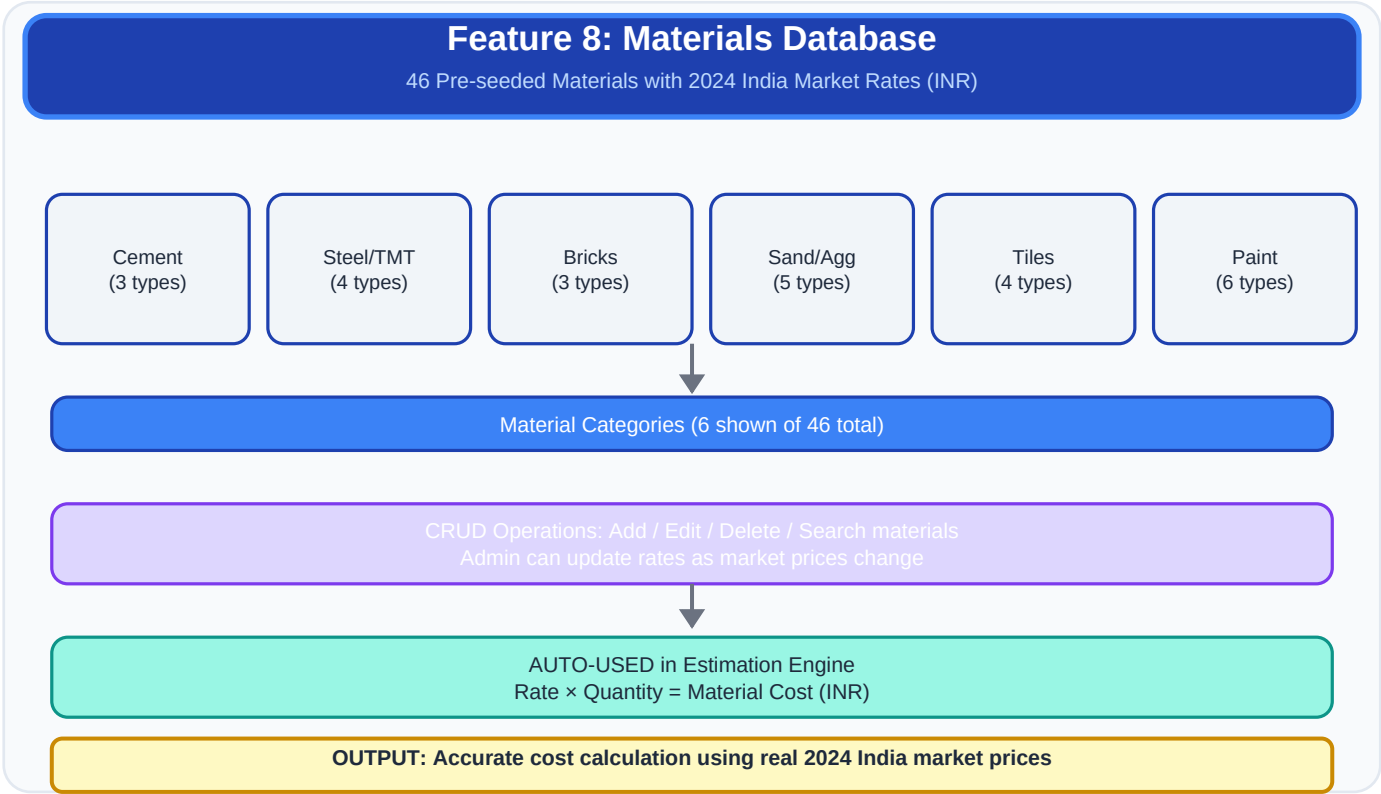
Create Project → Add Buildings → Add Rooms → Link to Estimates



Feature 7: Analytics Dashboard



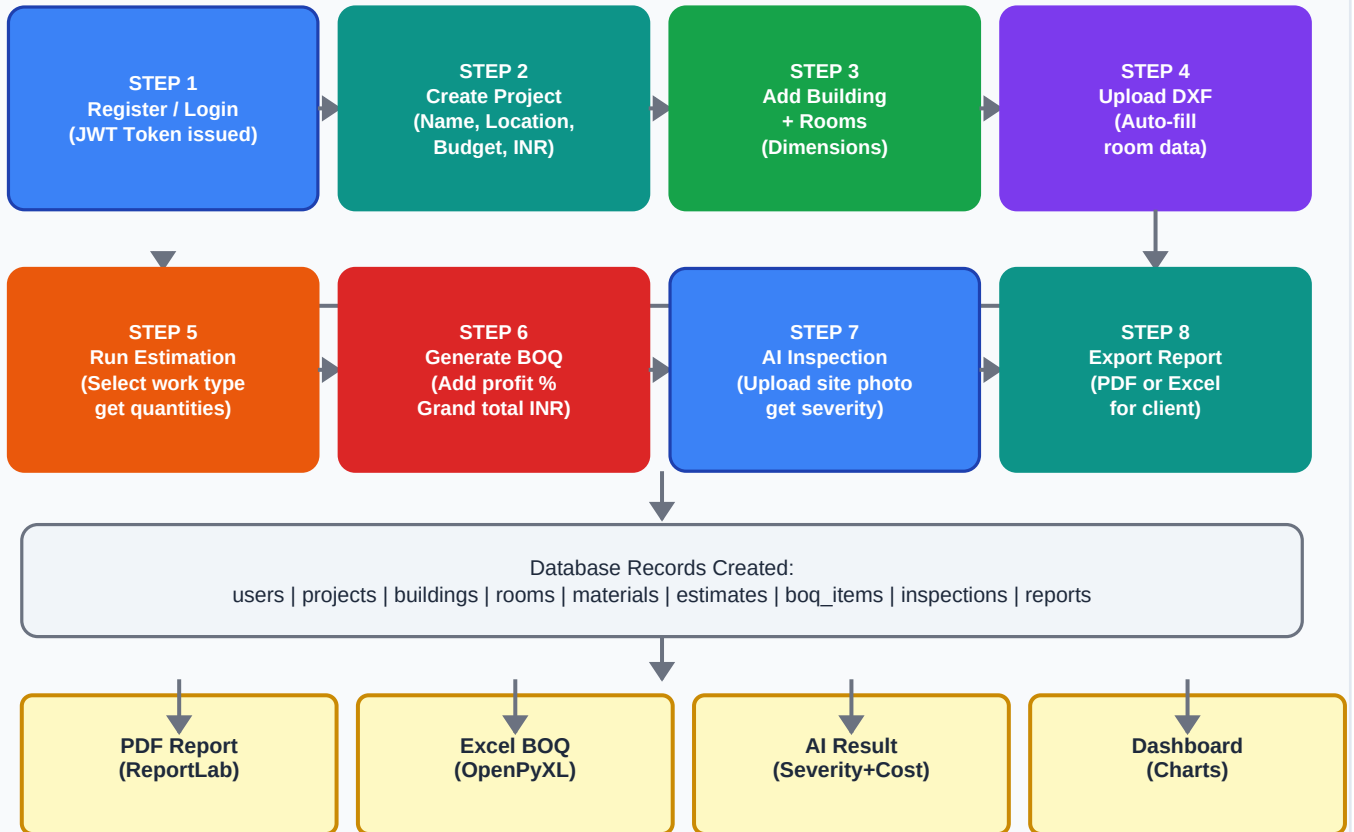
Feature 8: Materials Database



Complete End-to-End User Journey

Complete End-to-End User Journey

From Login to Final Report — Real Construction Scenario



Real Example: Site engineer in Bhopal uploads photo of RCC column
→ AI detects crack with 94% confidence → CRITICAL severity
→ Recommends grouting within 7 days → Cost: ₹20,000 – ₹80,000
→ Full PDF report generated and sent to structural consultant